MILITARY SPECIFICATION

ARMING WIRE ASSEMBLIES

1. SCOPE

1.1 Scope. This specification covers arming wire assemblies for use as arming elements for bomb fuzes and igniters.

1.2 Classification. Arming wire assemblies, shown on Drawing E22-128-1, shall be the type specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat.
TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bag and Sleeve, Interior Packaging.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

E22-128-1	-	Arming Wire, Assemblies, Assembly and Detail.
B13-19-51	-	Tester, Dynamic Load and Stripping Force, Arming Wires, Q57.
D150-2-1	-	Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing E22-128-1 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing E22-128-1 and subsidiary drawings.

3.2 Assembly. The arming wire shall be assembled as specified on Drawing E22-128-1 and subsidiary drawings.

3.3 Strength.

3.3.1 Swivel and loop assembly. The swivel and loop assembly (Drawing A14-5-1006 or Drawing C14-23-34) shall not distort or break when tested as specified in 4.4.4.1.

3.3.2 Arming wire assembly. The arming wire assembly shall not distort or break when tested as specified in 4.4.4.2.

3.4 Safety clip stripping force. When tested as specified in 4.4.4.3 the safety clip (Drawing B14-23-44) shall satisfy the following requirements:

(a) The force needed to strip the clip from the thicker wire shall be between 2-1/2 and 9 pounds.

(b) The force needed to strip the clip from the thinner wire shall be between 1 and 6 pounds.

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3.5 Swivel and loop assembly corrosion resistance. The loop and swivel assembly shall withstand the salt spray test as specified in 4.4.4.4.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of arming wire assemblies shall be produced in accordance with this specification for examination and test (see 4.3).

3.7 Workmanship. The arming wire assembly shall be free from foreign matter, (oil and grease) and damage such as kinks, bends, burrs or any deformation which would impair its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 280 arming wire assemblies shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

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4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105, general inspection level III.

4.3.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, using an AQL of 0.40; a separate sample shall be taken for salt spray test.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Sample assemblies shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For destructive tests. Sample assemblies shall be tested in accordance with 4.4.4 and MIL-STD-105, using an AQL of 0.40 percent defective.

4.3.4 Acceptance rejection criteria. The preproduction lot shall meet the examinations and tests specified herein to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the arming wire assemblies of one interfix, number produced by one manufacturer, during one week production under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.1.2 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.1 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105 using inspection level S-4 and an AQL of 0.40; a separate sample shall be taken for the salt spray test.

4.4.3 Inspection provisions.

4.4.3.1 For examination. Sample arming wire assemblies shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. The sample arming wire assemblies shall be tested as specified in 4.4.4 and MIL-STD-105 using an AQL of 0.40 percent defective.

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4.4.3.3 Classification of defects.

(a) Wire, arming - assembly (Drawing E22-128-1).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Component missing, incorrect or incorrectly assembled
102		Loop size incorrect
103		Wire length (min) incorrect
104		Swivel and loop assembly inoperative
105		Workmanship (3.7)
Minor:	AQL	4.0 percent defective
201		Wire improperly bent at sleeve

(b) Preparation for delivery (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL	2.5 percent defective
101		Packages or containers damaged
102		Unit package not as specified
103		Wood not preserved
104		Container closure incorrect
105		Strapping loose or missing
106		Marking missing, illegible or incorrect

4.4.4 Tests. The tests specified in 4.4.4.1, 4.4.4.2 and 4.4.4.3 shall be performed on the same assembly.

4.4.4.1 Strength of swivel and loop assembly. The swivel and loop assembly shall be subjected to a pull test by gradually and steadily applying a load up to 100 pounds and holding this load for a minimum of five seconds. The large loop of the assembly shall be supported by means of a hardened steel pin 9/32-inch square having the corners rounded on the flat upper surface against which the large loop bears. The load shall be

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applied by means of a hook or pin through the small loop. The assembly shall then be examined for distortion, cracks, or impaired condition of the swaged swivel head.

4.4.4.2 Strength of wire assembly. The end of the arming wire assembly shall be secured in the Tester, Dynamic Load and Stripping Force, Arming Wire, Q57 and the swivel loop attached to a hook whose radius is equal to that of the loop. The 5-pound weight shall be raised a distance of 20 inches above the extended limit of the wire and dropped in such a manner as to prevent kinking of the wire. The assembly shall be examined for distortion and damage.

4.4.4.3 Safety clip stripping force. The clips shall be tested at least 8 hours after assembling (one at a time) using the Tester, Dynamic Load and Stripping Force, Arming Wire, Q57. When a load of 2-1/2 pounds is applied the clip shall not strip from the wire. When a load of 9 pounds is applied the clip shall strip from the wire. The same procedure shall be followed with a 0.036 inch diameter wire, using weights of 1 - pound and 6 - pounds.

4.4.4.4 Salt spray. The loop and swivel assembly shall not corrode when tested in accordance with method 811 of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 Level A. The arming wire assemblies as shown on Drawing E22-128-1, shall be coiled, tied and individually placed in a foldable shipping card in the manner shown on the same drawing. The foldable shipping card shall be fabricated as shown on Drawing C14-28-383. Each card containing an arming wire assembly shall be sealed in a barrier bag conforming to type II, class E, style I or III, MIL-B-117, having inside dimensions as shown on Drawing C14-23-404. Included air volume shall be held to a practicable minimum. Sealed packages shall be capable of passing the applicable leakage tests for method IA-8, MIL-P-116.

5.1.2 Level B. Arming wire assemblies shall be packaged in accordance with 5.1.1.

5.1.3 Level C. Arming wire assemblies shall be packaged in accordance with Drawing E22-128-1 except enclosing the wire and folder assembly in a barrier bag or envelope is not required.

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5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Arming wire assemblies, packaged in accordance with 5.1.1, shall be packed, 100 per box, in a wood box conforming to class 2, style 4, for type 3 load with a 50 lb load limit, PPP-B-621. Inside dimensions of the box shall have approximate 25-1/4 by 17-1/2 by 9 inches length, width and depth, respectively. Boxes shall be preservative-heated by immersing all wood material in preservative conforming to type II, composition A, TT-W-572 for a minimum period of 3 minutes. Boxes shall not be loaded or marked until at least 24 hours after preservative treatment. Arrangement of the packages in the box shall be two rows of 50 each, on edge. Filler pads of weather-resistant fiberboard shall be included, at least one at each interior face, or necessary, to restrict free movement of the packages. Boxes shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to class B, type I or IV, QQ-S-781.

5.2.2 Level B. Arming wire assemblies, packaged in accordance with 5.1.1, shall be packed in accordance with 5.2.1 except wood preservative treatment of boxes is not required and strapping may be class A, QQ-S-781.

5.2.3 Level C. Arming wire assemblies packaged as specified (see 6.2) shall be packed in fiberboard boxes conforming to weather-resistant class, PPP-B-636. Quantity per box shall be as specified in the contract or order or in even unit quantities not to exceed the size and weight limations of the respective box in PPP-B-636. Packing shall include filler pads, as necessary, to restrict free movement of the packages. Boxes shall be closed by gluing and strapped, in accordance with the appendix to PPP-B-636.

5.3 Marking. In addiiton to any special marking required in the contract or order, marking shall be as specified herein.

5.3.1 Package marking. Each package shall be marked with item identification marking, including the Federal stock number, in accordance with Drawing E22-128-1.

5.3.2 Shipping container marking (exterior box). Each box shall be marked in accordance with Drawing D150-2-1, as applicable.

5.4 Palletization. When specified in the contract or order (mandatory for level A (see 6.2), except when quantity is less than pallet load) arming wires, packed as specified, shall be palletized and the unit load marked in accordance with Drawing D14-23-1884.

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6. NOTES

6.1 Intended use. This specification covers arming wire assemblies used to prevent arming of the fuze while the bomb is being flown to the release point. They are single or double wires, depending on the fuzing system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required.
(c) Whether or not palletization is required.
(d) The type of arming wire assembly required.
(e) Tester Q57 will be Government furnished (GFE).
(f) Preproduction.

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1325-A312

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